



**Generation Interconnection
System Impact Study Report
for
Queue Project AF2-115
JETERSVILLE-PONTON 115 KV
15 MW Capacity / 25 MW Energy**

February 2021

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Dominion.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC) has proposed an uprate to a planned Solar generating facility located in Amelia, Virginia. This project is an increase to the Interconnection Customer's AF1-294 project, which will share the same point of interconnection. The AF2-115 queue position is a 25 MW uprate (15 MW Capacity uprate) to the previous project. The total installed facilities will have a capability of 66 MW with 39.6 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this uprate project is December 01, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-115
Project Name	JETERSVILLE-PONTON 115 KV
State	Virginia
County	Amelia
Transmission Owner	Dominion
MFO	66
MWE	25
MWC	15
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF2-115 will interconnect with the Dominion on transmission system as an uprate to AF1-294 which is tapping the Jetersville to Ponton 115 kV line.

5 Cost Summary

The AF2-115 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 0
Total System Network Upgrade Costs (Summer Peak)	\$ 1,708,558
Total Costs	\$ 1,708,558

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

Note 3: If the AF1-294 project withdraws from the New Services Queue, then the AF2-115 customer will be responsible for the interconnection facilities described in the AF1-294 Facilities Study Agreement.

6 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AF2-115 was evaluated as a 15 MW Capacity (25.0 MW Energy) injection at the AF1-294 115 kV substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2023 AF2 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically, in Planning Studies, NERC Planning Event 3 and 6 Contingency Conditions (Loss of generator, transmission circuit, transformer, shunt device, or Single Pole of a DC line followed by the loss of a generator, transmission circuit, transformer, shunt device or single pole of a DC line) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

The physical interconnection work identified in the AF1-294 project reports is sufficient to support the additional generation proposed under the AF2-115 project.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Total Physical Interconnection Costs	\$ 0

7 Schedule

The schedule for any required Network Impact Reinforcements will be more clearly identified in future study phases. The estimated time to complete each of the required reinforcements is identified in the "System Reinforcements" section of the report.

8 Transmission Owner Analysis

8.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2023 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no further deficiencies were identified.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion’s “Dominion Energy Electric Transmission Generator Interconnection Requirements” documented in Dominion’s Facility Interconnection Requirements “Exhibit C” located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC’s GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with

these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Irradiance (Watts/meter²) - (Required for plants with Maximum Facility Output of 3 MW or higher)
- Ambient air temperature (Fahrenheit) - (Accepted, not required)
- Wind speed (meters/second) - (Accepted, not required)
- Wind direction (decimal degrees from true north) - (Accepted, not required)

10.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

11 Summer Peak Analysis

The Queue Project AF2-115 was evaluated as a 25.0 MW (Capacity 15.0 MW) injection as an uprate to AF1-294 which is tapping the Jetersville to Ponton 115 kV line in the Dominion area. Project AF2-115 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-115 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
97733461	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P4-2: 511T591	breaker	699.0	126.35	127.68	AC	10.08
97733463	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P4-2: 511T556	breaker	699.0	126.35	127.68	AC	10.08
106270498	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P1-2: LN 556-B	single	571.52	110.75	111.8	AC	6.04
157954181	314677	6BUCKING	230.0	DVP	314747	6BREMO	230.0	DVP	1	DVP_P4-2: 556T591-A	breaker	699.0	126.35	127.67	AC	10.08
97733476	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P4-2: 511T591	breaker	684.0	125.45	126.8	AC	10.08
97733477	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P4-2: 511T556	breaker	684.0	125.45	126.8	AC	10.08
106270509	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P1-2: LN 556-B	single	559.3	110.29	111.36	AC	6.04
157954186	314692	6FARMVIL	230.0	DVP	314677	6BUCKING	230.0	DVP	1	DVP_P4-2: 556T591-A	breaker	684.0	125.44	126.8	AC	10.08
96866983	314702	3KERR	115.0	DVP	304102	3GW KING TAP	115.0	CPL	1	DVP_P4-6: CAROLIN T122	breaker	199.0	119.81	121.06	AC	3.02

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPAC T
96867504	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P7-1: LN 22-90-A	tower	199.0	125.16	126.4	AC	2.98
96867505	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P7-1: LN 22-90-B	tower	199.0	115.31	116.55	AC	2.98
11933625 1	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 2202	breaker	199.0	111.13	112.27	AC	2.64
11933625 2	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 101002	breaker	199.0	108.46	109.6	AC	2.64
11933625 3	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 102802	breaker	199.0	108.46	109.6	AC	2.64
14536528 9	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P4-2: 13002	breaker	199.0	108.46	109.6	AC	2.64

11.4 Steady-State Voltage Requirements

To be determined at Facilities Study phase.

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPAC T
97942975	31386 7	6BREMODIST	230.0	DVP	31370 7	6FORK UNION	230.0	DVP	1	DVP_P 1-2: LN 2027-B	operation	661.76	127.53	128.52	AC	6.75
10627049 5	31467 7	6BUCKING	230.0	DVP	31474 7	6BREMO	230.0	DVP	1	DVP_P 1-2: LN 556-B	operation	571.52	150.49	152.27	AC	10.07
10627050 6	31469 2	6FARMVIL	230.0	DVP	31467 7	6BUCKING	230.0	DVP	1	DVP_P 1-2: LN 556-B	operation	559.3	149.36	151.18	AC	10.07
97204531	31470 1	3LONEPN	115.0	DVP	31470 7	3MORAN	115.0	DVP	1	DVP_P 1-2: LN 98-C	operation	203.98	102.79	114.55	AC	24.99
15855176 4	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P 1-2: LN 296-B	operation	199.0	104.51	105.54	AC	2.45
15855176 5	31470 2	3KERR	115.0	DVP	30410 2	3GW KING TAP	115.0	CPL	1	DVP_P 1-2: LN 296-A	operation	199.0	104.51	105.53	AC	2.45
97204547	31470 7	3MORAN	115.0	DVP	31469 1	3FARMVIL	115.0	DVP	1	DVP_P 1-2: LN 98-C	operation	203.98	100.74	112.42	AC	24.99

ID	FROM BUS#	FROM BUS	kV	FRO M BUS AREA	TO BUS#	TO BUS	kV	TO BUS ARE A	CK T ID	CONT NAME	Type	Ratin g MVA	PRE PROJEC T LOADIN G %	POST PROJEC T LOADIN G %	AC D C	MW IMPAC T
97942955	31474 7	6BREMO	230. 0	DVP	31386 7	6BREMODI ST	230. 0	DVP	1	DVP_P 1-2: LN 2027-B	operatio n	661.7 6	129.84	130.83	AC	6.75

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-115	Upgrade Number																																				
157954186,977 33476,9773347 7,106270509	2	6FARMVIL 230.0 kV - 6BUCKING 230.0 kV Ckt 1	DVP ProjectId : n6245 (dom-035) Description : Rebuild 12.76 miles of 230 kV Line 298 from Buckingham to Farmville with 2-636 ACSR. Type : FAC Total Cost : \$19,140,000 Time Estimate : 30-36 Months Ratings : 1047.0/1047.0/1204.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-291</td><td>5.07</td><td>2.12%</td><td>\$406,637</td></tr><tr><td>AE2-292</td><td>13.87</td><td>5.81%</td><td>\$1,112,436</td></tr><tr><td>AF1-059</td><td>5.01</td><td>2.10%</td><td>\$401,825</td></tr><tr><td>AF1-246</td><td>29.51</td><td>12.37%</td><td>\$2,366,835</td></tr><tr><td>AF1-293</td><td>11.03</td><td>4.62%</td><td>\$884,656</td></tr><tr><td>AF1-294</td><td>16.53</td><td>6.93%</td><td>\$1,325,780</td></tr><tr><td>AF2-042</td><td>147.54</td><td>61.83%</td><td>\$11,833,371</td></tr><tr><td>AF2-115</td><td>10.08</td><td>4.22%</td><td>\$808,461</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-291	5.07	2.12%	\$406,637	AE2-292	13.87	5.81%	\$1,112,436	AF1-059	5.01	2.10%	\$401,825	AF1-246	29.51	12.37%	\$2,366,835	AF1-293	11.03	4.62%	\$884,656	AF1-294	16.53	6.93%	\$1,325,780	AF2-042	147.54	61.83%	\$11,833,371	AF2-115	10.08	4.22%	\$808,461	\$19,140,000	\$808,461	n6245
Queue	MW	Cost %	Cost \$																																							
AE2-291	5.07	2.12%	\$406,637																																							
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AF2-115	10.08	4.22%	\$808,461																																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-115	Upgrade Number
145365289,968 66983,9686750 5,96867504,119 336253,119336 252,119336251	3	3KERR 115.0 kV - 3GW KING TAP 115.0 kV Ckt 1	DVP ProjectId : n6115 (dom-002) Description : For DEV portion, rebuild 4.7 miles of 115 kV Line 45 from Kerr Dam to GW King Tap with 768 ACSS. Type : FAC Total Cost : \$6,123,000 Time Estimate : 30-36 Months Ratings : 393.0/393.0/452.0 Queue Project AF2-115 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF2-115 could receive cost allocation. Note 2: Although Queue Project AF2-115 may not have cost responsibility for this upgrade, Queue Project AF2-115 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-115 comes into service prior to completion of the upgrade, Queue Project AF2-115 will need an interim study DEP ProjectId : dep0002 Description : Reconductor 20 miles with single 795 ACSS-TW per phase, replace breaker, disconnect switches, wave trap, line switches. Type : FAC Total Cost : \$80,000,000 Time Estimate : 48 Months Ratings : 301.0/301.0/301.0 Queue Project AF2-115 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF2-115 could receive cost allocation. Note 2: Although Queue Project AF2-115 may not have cost responsibility for this upgrade, Queue Project AF2-115 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF2-115 comes into service prior to completion of the upgrade, Queue Project AF2-115 will need an interim study	\$86,123,000	\$0	n6115

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-115	Upgrade Number																																								
106270498,977 33461,1579541 81,97733463	1	6BUCKING 230.0 kV - 6BREMO 230.0 kV Ckt 1	DVP ProjectId : n6219 (dom-034) Description : Rebuild 15.51 miles of 230 kV Line 298 from Buckingham to Brema with 2-636 ACSR. Type : FAC Total Cost : \$23,265,000 Time Estimate : 30-36 Months Ratings : 1047.0/1047.0/1204.0 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-259</td><td>5.2</td><td>2.00%</td><td>\$464,336</td></tr><tr><td>AE2-291</td><td>11.14</td><td>4.28%</td><td>\$994,750</td></tr><tr><td>AE2-292</td><td>13.87</td><td>5.32%</td><td>\$1,238,526</td></tr><tr><td>AF1-059</td><td>15.67</td><td>6.01%</td><td>\$1,399,258</td></tr><tr><td>AF1-246</td><td>29.5</td><td>11.32%</td><td>\$2,634,212</td></tr><tr><td>AF1-293</td><td>11.04</td><td>4.24%</td><td>\$985,820</td></tr><tr><td>AF1-294</td><td>16.53</td><td>6.34%</td><td>\$1,476,051</td></tr><tr><td>AF2-042</td><td>147.51</td><td>56.62%</td><td>\$13,171,951</td></tr><tr><td>AF2-115</td><td>10.08</td><td>3.87%</td><td>\$900,097</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-259	5.2	2.00%	\$464,336	AE2-291	11.14	4.28%	\$994,750	AE2-292	13.87	5.32%	\$1,238,526	AF1-059	15.67	6.01%	\$1,399,258	AF1-246	29.5	11.32%	\$2,634,212	AF1-293	11.04	4.24%	\$985,820	AF1-294	16.53	6.34%	\$1,476,051	AF2-042	147.51	56.62%	\$13,171,951	AF2-115	10.08	3.87%	\$900,097	\$23,265,000	\$900,097	n6219
Queue	MW	Cost %	Cost \$																																											
AE2-259	5.2	2.00%	\$464,336																																											
AE2-291	11.14	4.28%	\$994,750																																											
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AF1-293	11.04	4.24%	\$985,820																																											
AF1-294	16.53	6.34%	\$1,476,051																																											
AF2-042	147.51	56.62%	\$13,171,951																																											
AF2-115	10.08	3.87%	\$900,097																																											
			TOTAL COST	\$128,528,000	\$1,708,558																																									

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
157954181	314677	6BUCKING	DVP	314747	6BREMO	DVP	1	DVP_P4-2: 556T591-A	breaker	699.0	126.35	127.67	AC	10.08

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.2168	50/50	0.2168
314429	3JTRSVLE	0.8671	50/50	0.8671
314677	6BUCKING	1.5835	50/50	1.5835
315153	1CLOVER1	21.8880	50/50	21.8880
315154	1CLOVER2	21.6699	50/50	21.6699
924022	AB2-043 E O1	3.2833	Adder	3.86
924032	AB2-045 E	4.3976	50/50	4.3976
924162	AB2-060 E OP	5.0964	Adder	6.0
924301	AB2-077 C O1 (Suspended)	2.2891	Adder	2.69
924302	AB2-077 E O1 (Suspended)	1.5261	Adder	1.8
924311	AB2-078 C O1 (Suspended)	2.2891	Adder	2.69
924312	AB2-078 E O1 (Suspended)	1.5261	Adder	1.8
924321	AB2-079 C O1 (Suspended)	2.2891	Adder	2.69
924322	AB2-079 E O1 (Suspended)	1.5261	Adder	1.8
925611	AC1-036 C	1.8364	50/50	1.8364
925612	AC1-036 E	2.9963	50/50	2.9963
925781	AC1-054 C O1	5.1726	Adder	6.09
925782	AC1-054 E O1	2.3829	Adder	2.8
925831	AC1-062	0.1287	50/50	0.1287
925991	AC1-075 C	4.2230	Adder	4.97
925992	AC1-075 E	2.3927	Adder	2.81
926021	AC1-080 C	1.4114	Adder	1.66
926022	AC1-080 E	0.7939	Adder	0.93
926271	AC1-105 C O1	5.7572	Adder	6.77
926272	AC1-105 E O1	2.8659	Adder	3.37
926451	AC1-116 C	-0.3879	Adder	-0.46
927251	AC1-221 C	1.4232	Adder	1.67
927252	AC1-221 E	1.4232	Adder	1.67
927261	AC1-222 C	3.4595	Adder	4.07
927262	AC1-222 E	3.2933	Adder	3.87
932511	AC2-071 C (Withdrawn : 01/22/2021)	9.2838	50/50	9.2838
932512	AC2-071 E (Withdrawn : 01/22/2021)	4.6768	50/50	4.6768
932761	AC2-100 C	3.2753	Adder	3.85

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
932762	AC2-100 E	1.5986	Adder	1.88
934311	AD1-055 C	2.4020	Adder	2.83
934312	AD1-055 E	0.6194	Adder	0.73
934341	AD1-058 C	3.5677	Adder	4.2
934342	AD1-058 E	0.9065	Adder	1.07
934611	AD1-087 C O1	10.2498	Adder	12.06
934612	AD1-087 E O1	4.8172	Adder	5.67
934621	AD1-088 C	35.2780	50/50	35.2780
934622	AD1-088 E	16.6014	50/50	16.6014
934991	AD1-131 C	1.1697	Adder	1.38
934992	AD1-131 E	0.7798	Adder	0.92
935171	AD1-152 C O1	10.1861	Adder	11.98
935172	AD1-152 E O1	6.7908	Adder	7.99
935221	AD1-157 C (Withdrawn : 01/26/2021)	3.8761	50/50	3.8761
935222	AD1-157 E (Withdrawn : 01/26/2021)	2.5841	50/50	2.5841
936261	AD2-033 C	23.8095	50/50	23.8095
936262	AD2-033 E	15.8730	50/50	15.8730
936361	AD2-046 C O1	7.9885	Adder	9.4
936362	AD2-046 E O1	3.6735	Adder	4.32
936481	AD2-063 C O1	20.2442	Adder	23.82
936482	AD2-063 E O1	13.3837	Adder	15.75
937481	AD2-202 C O1	2.7163	Adder	3.2
937482	AD2-202 E O1	1.5279	Adder	1.8
938371	AE1-056 C	16.7104	50/50	16.7104
938372	AE1-056 E	9.1304	50/50	9.1304
939181	AE1-148 C O1	8.0059	Adder	9.42
939182	AE1-148 E O1	5.3373	Adder	6.28
940241	AE2-006	0.9343	50/50	0.9343
940661	AE2-053 O1	2.9651	Adder	3.49
941791	AE2-182 C	5.7314	50/50	5.7314
941792	AE2-182 E	2.6681	50/50	2.6681
942451	AE2-258	2.7393	Adder	3.22
942461	AE2-259 C O1	30.4836	50/50	30.4836
942462	AE2-259 E O1	20.3224	50/50	20.3224
942751	AE2-291 C O1	5.6723	Adder	6.67
942752	AE2-291 E O1	3.7815	Adder	4.45
942761	AE2-292 C O1	7.0625	Adder	8.31
942762	AE2-292 E O1	4.7083	Adder	5.54
943901	AF1-058 C	1.9500	Adder	2.29
943902	AF1-058 E	1.3000	Adder	1.53
943911	AF1-059	13.3237	Adder	15.67
945811	AF1-246 C O1	10.5319	Adder	12.39
945812	AF1-246 E O1	14.5440	Adder	17.11
946293	AF1-293 BAT	11.0398	Merchant Transmission	11.0398
946301	AF1-294 C	9.9170	50/50	9.9170
946302	AF1-294 E	6.6113	50/50	6.6113
957481	AF2-042 C	75.2276	Adder	88.5
957482	AF2-042 E	50.1517	Adder	59.0
958211	AF2-115 C	6.0470	50/50	6.0470
958212	AF2-115 E	4.0313	50/50	4.0313
958801	AF2-171 C	36.0585	50/50	36.0585

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
958802	AF2-171 E	24.0390	50/50	24.0390
959311	AF2-222 C	33.3610	50/50	33.3610
959312	AF2-222 E	22.3519	50/50	22.3519
959751	AF2-266	12.5388	Adder	14.75
960061	AF2-297 C	7.8001	Adder	9.18
960062	AF2-297 E	5.2001	Adder	6.12
961063	AF2-397 BAT	7.7164	Merchant Transmission	7.7164
WEC	WEC	0.2977	Confirmed LTF	0.2977
LGEE	LGEE	0.5625	Confirmed LTF	0.5625
CPLE	CPLE	3.4947	Confirmed LTF	3.4947
CBM-W2	CBM-W2	12.4079	Confirmed LTF	12.4079
NY	NY	0.3888	Confirmed LTF	0.3888
TVA	TVA	2.6446	Confirmed LTF	2.6446
O-066	O-066	5.3962	Confirmed LTF	5.3962
CBM-S2	CBM-S2	23.4495	Confirmed LTF	23.4495
CBM-S1	CBM-S1	14.6629	Confirmed LTF	14.6629
G-007	G-007	0.8414	Confirmed LTF	0.8414
MADISON	MADISON	0.7822	Confirmed LTF	0.7822
MEC	MEC	1.8353	Confirmed LTF	1.8353
AA2-074	AA2-074	2.3810	LTF	2.3810
CBM-W1	CBM-W1	11.0588	Confirmed LTF	11.0588

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
157954186	314692	6FARMVIL	DVP	314677	6BUCKING	DVP	1	DVP_P4-2: 556T591-A	breaker	684.0	125.44	126.8	AC	10.08

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
313853	3PONTONDP	0.2168	50/50	0.2168
314429	3JTRSVLE	0.8672	50/50	0.8672
315153	1CLOVER1	21.8932	50/50	21.8932
315154	1CLOVER2	21.6750	50/50	21.6750
924022	AB2-043 E O1	3.2843	Adder	3.86
924162	AB2-060 E OP	5.0979	Adder	6.0
924301	AB2-077 C O1 (Suspended)	2.2898	Adder	2.69
924302	AB2-077 E O1 (Suspended)	1.5265	Adder	1.8
924311	AB2-078 C O1 (Suspended)	2.2898	Adder	2.69
924312	AB2-078 E O1 (Suspended)	1.5265	Adder	1.8
924321	AB2-079 C O1 (Suspended)	2.2898	Adder	2.69

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
924322	AB2-079 E O1 (Suspended)	1.5265	Adder	1.8
925611	AC1-036 C	1.8368	50/50	1.8368
925612	AC1-036 E	2.9969	50/50	2.9969
925781	AC1-054 C O1	5.1752	Adder	6.09
925782	AC1-054 E O1	2.3841	Adder	2.8
925831	AC1-062	0.1287	50/50	0.1287
925991	AC1-075 C	4.2253	Adder	4.97
925992	AC1-075 E	2.3940	Adder	2.82
926021	AC1-080 C	1.4121	Adder	1.66
926022	AC1-080 E	0.7943	Adder	0.93
926271	AC1-105 C O1	5.7593	Adder	6.78
926272	AC1-105 E O1	2.8669	Adder	3.37
926451	AC1-116 C	-0.3876	Adder	-0.46
927251	AC1-221 C	1.4240	Adder	1.68
927252	AC1-221 E	1.4240	Adder	1.68
927261	AC1-222 C	3.4609	Adder	4.07
927262	AC1-222 E	3.2946	Adder	3.88
932761	AC2-100 C	3.2773	Adder	3.86
932762	AC2-100 E	1.5996	Adder	1.88
934311	AD1-055 C	2.4030	Adder	2.83
934312	AD1-055 E	0.6196	Adder	0.73
934341	AD1-058 C	3.5699	Adder	4.2
934342	AD1-058 E	0.9071	Adder	1.07
934611	AD1-087 C O1	10.2527	Adder	12.06
934612	AD1-087 E O1	4.8185	Adder	5.67
934621	AD1-088 C	35.2830	50/50	35.2830
934622	AD1-088 E	16.6038	50/50	16.6038
934991	AD1-131 C	1.1704	Adder	1.38
934992	AD1-131 E	0.7803	Adder	0.92
935171	AD1-152 C O1	10.1890	Adder	11.99
935172	AD1-152 E O1	6.7927	Adder	7.99
935221	AD1-157 C (Withdrawn : 01/26/2021)	3.8768	50/50	3.8768
935222	AD1-157 E (Withdrawn : 01/26/2021)	2.5846	50/50	2.5846
936261	AD2-033 C	23.8150	50/50	23.8150
936262	AD2-033 E	15.8766	50/50	15.8766
936361	AD2-046 C O1	7.9917	Adder	9.4
936362	AD2-046 E O1	3.6750	Adder	4.32
936481	AD2-063 C O1	20.2495	Adder	23.82
936482	AD2-063 E O1	13.3872	Adder	15.75
937481	AD2-202 C O1	2.7171	Adder	3.2
937482	AD2-202 E O1	1.5283	Adder	1.8
938371	AE1-056 C	16.7135	50/50	16.7135
938372	AE1-056 E	9.1321	50/50	9.1321
938561	AE1-075 C	-0.9177	Adder	-1.08
939181	AE1-148 C O1	8.0091	Adder	9.42
939182	AE1-148 E O1	5.3394	Adder	6.28
940241	AE2-006	0.9345	50/50	0.9345
940661	AE2-053 O1	2.9663	Adder	3.49
941791	AE2-182 C	5.7323	50/50	5.7323
941792	AE2-182 E	2.6685	50/50	2.6685

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
942451	AE2-258	2.7401	Adder	3.22
942461	AE2-259 C O1	30.4878	50/50	30.4878
942462	AE2-259 E O1	20.3252	50/50	20.3252
942751	AE2-291 C O1	5.6759	Adder	6.68
942752	AE2-291 E O1	3.7839	Adder	4.45
942761	AE2-292 C O1	7.0671	Adder	8.31
942762	AE2-292 E O1	4.7114	Adder	5.54
943901	AF1-058 C	1.9507	Adder	2.29
943902	AF1-058 E	1.3005	Adder	1.53
943911	AF1-059	13.3297	Adder	15.68
945811	AF1-246 C O1	10.5344	Adder	12.39
945812	AF1-246 E O1	14.5474	Adder	17.11
946293	AF1-293 BAT	11.0341	Merchant Transmission	11.0341
946301	AF1-294 C	9.9187	50/50	9.9187
946302	AF1-294 E	6.6125	50/50	6.6125
957481	AF2-042 C	75.2454	Adder	88.52
957482	AF2-042 E	50.1636	Adder	59.02
958211	AF2-115 C	6.0480	50/50	6.0480
958212	AF2-115 E	4.0320	50/50	4.0320
958503	AF2-144 BAT	0.8923	Merchant Transmission	0.8923
958801	AF2-171 C	36.0648	50/50	36.0648
958802	AF2-171 E	24.0432	50/50	24.0432
959311	AF2-222 C	33.3680	50/50	33.3680
959312	AF2-222 E	22.3566	50/50	22.3566
959751	AF2-266	12.5418	Adder	14.76
960061	AF2-297 C	7.8030	Adder	9.18
960062	AF2-297 E	5.2020	Adder	6.12
961063	AF2-397 BAT	7.7127	Merchant Transmission	7.7127
WEC	WEC	0.2999	Confirmed LTF	0.2999
LGEE	LGEE	0.5665	Confirmed LTF	0.5665
CPL	CPL	3.4993	Confirmed LTF	3.4993
CBM-W2	CBM-W2	12.4652	Confirmed LTF	12.4652
NY	NY	0.3843	Confirmed LTF	0.3843
TVA	TVA	2.6544	Confirmed LTF	2.6544
O-066	O-066	5.3491	Confirmed LTF	5.3491
CBM-S2	CBM-S2	23.4899	Confirmed LTF	23.4899
CBM-S1	CBM-S1	14.7226	Confirmed LTF	14.7226
G-007	G-007	0.8341	Confirmed LTF	0.8341
MADISON	MADISON	0.7822	Confirmed LTF	0.7822
MEC	MEC	1.8464	Confirmed LTF	1.8464
AA2-074	AA2-074	2.3846	LTF	2.3846
CBM-W1	CBM-W1	11.1464	Confirmed LTF	11.1464

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96867504	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P7-1: LN 22-90-A	tower	199.0	125.16	126.4	AC	2.98

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
315158	1KERR 1	0.8811	50/50	0.8811
315159	1KERR 2	2.4670	50/50	2.4670
315160	1KERR 3	2.4670	50/50	2.4670
315161	1KERR 4	2.4670	50/50	2.4670
315162	1KERR 5	2.4670	50/50	2.4670
315163	1KERR 6	2.4670	50/50	2.4670
315164	1KERR 7	2.4670	50/50	2.4670
924021	AB2-043 C O1	0.6180	50/50	0.6180
924022	AB2-043 E O1	3.2884	50/50	3.2884
924161	AB2-060 C OP	1.7564	50/50	1.7564
924162	AB2-060 E OP	4.9188	50/50	4.9188
924301	AB2-077 C O1 (Suspended)	2.3768	50/50	2.3768
924302	AB2-077 E O1 (Suspended)	1.5846	50/50	1.5846
924311	AB2-078 C O1 (Suspended)	2.3768	50/50	2.3768
924312	AB2-078 E O1 (Suspended)	1.5846	50/50	1.5846
924321	AB2-079 C O1 (Suspended)	2.3768	50/50	2.3768
924322	AB2-079 E O1 (Suspended)	1.5846	50/50	1.5846
925611	AC1-036 C	0.8793	50/50	0.8793
925612	AC1-036 E	1.4346	50/50	1.4346
925781	AC1-054 C O1	15.5550	50/50	15.5550
925782	AC1-054 E O1	7.1658	50/50	7.1658
926271	AC1-105 C O1	2.4425	Adder	2.87
926272	AC1-105 E O1	1.2158	Adder	1.43
935221	AD1-157 C (Withdrawn : 01/26/2021)	0.8225	Adder	0.97
935222	AD1-157 E (Withdrawn : 01/26/2021)	0.5484	Adder	0.65
936261	AD2-033 C	12.6001	50/50	12.6001
936262	AD2-033 E	8.4001	50/50	8.4001
936361	AD2-046 C O1	17.8270	50/50	17.8270
936362	AD2-046 E O1	8.1978	50/50	8.1978
936481	AD2-063 C O1	16.1145	50/50	16.1145
936482	AD2-063 E O1	10.6535	50/50	10.6535
938371	AE1-056 C	3.5460	Adder	4.17
938372	AE1-056 E	1.9375	Adder	2.28
939181	AE1-148 C O1	17.2071	50/50	17.2071

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939182	AE1-148 E O1	11.4714	50/50	11.4714
940241	AE2-006	0.4474	50/50	0.4474
940661	AE2-053 O1	6.3730	50/50	6.3730
942451	AE2-258	2.7436	50/50	2.7436
942461	AE2-259 C O1	3.7837	Adder	4.45
942462	AE2-259 E O1	2.5225	Adder	2.97
943901	AF1-058 C	0.5678	Adder	0.67
943902	AF1-058 E	0.3786	Adder	0.45
943911	AF1-059	12.3240	50/50	12.3240
946301	AF1-294 C	2.4962	Adder	2.94
946302	AF1-294 E	1.6642	Adder	1.96
958211	AF2-115 C	1.5221	Adder	1.79
958212	AF2-115 E	1.0147	Adder	1.19
958801	AF2-171 C	10.8414	50/50	10.8414
958802	AF2-171 E	7.2276	50/50	7.2276
959311	AF2-222 C	14.9340	50/50	14.9340
959312	AF2-222 E	10.0058	50/50	10.0058
960061	AF2-297 C	2.2713	Adder	2.67
960062	AF2-297 E	1.5142	Adder	1.78
NEWTON	NEWTON	0.4481	Confirmed LTF	0.4481
FARMERCITY	FARMERCITY	0.0260	Confirmed LTF	0.0260
G-007A	G-007A	0.1654	Confirmed LTF	0.1654
VFT	VFT	0.4386	Confirmed LTF	0.4386
GIBSON	GIBSON	0.2151	Confirmed LTF	0.2151
PRAIRIE	PRAIRIE	1.2243	Confirmed LTF	1.2243
COFFEEN	COFFEEN	0.0838	Confirmed LTF	0.0838
CHEOAH	CHEOAH	0.3814	Confirmed LTF	0.3814
EDWARDS	EDWARDS	0.1347	Confirmed LTF	0.1347
TILTON	TILTON	0.2375	Confirmed LTF	0.2375
CALDERWOOD	CALDERWOOD	0.3713	Confirmed LTF	0.3713
BLUEG	BLUEG	0.6597	Confirmed LTF	0.6597
TRIMBLE	TRIMBLE	0.2092	Confirmed LTF	0.2092
CATAWBA	CATAWBA	0.4193	Confirmed LTF	0.4193

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA2-074	CPL-PJM	Confirmed
AB2-043	Chase City 115kV	Under Construction
AB2-045	Buckingham 34.5kV	In Service
AB2-060	Chase City-Lunenburg 115kV	In Service
AB2-077	Buggs Island-Chase City 115kV	Suspended
AB2-078	Buggs Island-Chase City 115kV	Suspended
AB2-079	Buggs Island-Chase City 115kV	Suspended
AC1-036	Twittys Creek 34.5kV	Partially in Service - Under Construction
AC1-054	Kerr Dam-Eatons Ferry 115 kV	Engineering and Procurement
AC1-062	Jetersville 34.5kV	In Service
AC1-075	Perth-Hickory Grove 115kV	Under Construction
AC1-080	Perth-Hickory Grove 115kV	Under Construction
AC1-105	Halifax-Mt. Laurel 115kV	Engineering and Procurement
AC1-116	Mount Eagle 34.5kV	Engineering and Procurement
AC1-221	Halifax-Person 230kV	Engineering and Procurement
AC1-222	Crystal Hill-Halifax 115kV	Engineering and Procurement
AC2-071	Buckingham 35kV	Withdrawn
AC2-100	Halifax-Person 230kV	Engineering and Procurement
AD1-055	Crystal Hill-Halifax 115 kV	Engineering and Procurement
AD1-058	Halifax-Person 230 kV	Engineering and Procurement
AD1-087	Clover-Sedge Hill 230 kV	Active
AD1-088	Briery-Clover 230 kV	Active
AD1-131	Sedge Hill-Person 230 kV	Engineering and Procurement
AD1-152	Clover-Sedge Hill 230 kV	Active
AD1-157	South Creek 34.5 kV	Withdrawn
AD2-033	Chase City-Lunenburg 115 kV	Active
AD2-046	Boydton DP-Kerr Dam 115 kV	Active
AD2-063	Central-Chase City 115kV	Active
AD2-202	Clover-Sedge Hill 230kV	Active
AE1-056	Red House-South Creek 115 kV	Active
AE1-075	Powhatan 34.5 kV	Engineering and Procurement
AE1-148	Kerr Dam-Ridge Rd 115 kV	Active
AE2-006	Twittys Creek 34.5 kV	Partially in Service - Under Construction
AE2-053	Kerr Dam-Ridge Road 115 kV	Active
AE2-182	Briery-Clover 230 kV	Active
AE2-258	Chase City 115 kV	Active
AE2-259	Curdsville-Willis Mtn 115 kV	Active
AE2-291	Grit DP-Perth 115 kV	Active
AE2-292	Grit DP-Perth 115 kV	Active

Queue Number	Project Name	Status
AF1-058	Welco 34.5 kV	Engineering and Procurement
AF1-059	Brodnax-South Hill 115 kV	Active
AF1-246	Clover-Rawlings 500 kV	Active
AF1-293	Kidds Store-Fort Union 115 kV	Active
AF1-294	Jetersville-Ponton 115 kV	Active
AF2-042	Clover-Rawlings 500 kV	Active
AF2-115	Jetersville-Ponton 115 kV	Active
AF2-144	Powhatan 34.5 kV	Active
AF2-171	Madisonville 115 kV	Active
AF2-222	Madisonville DP-Twitty's Creek 115 kV	Active
AF2-266	Clover 230 kV	Active
AF2-297	Sedge Hill 115 kV	Active
AF2-397	Fork Union-Mt. Eagle 230 kV	Active

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P4-2: 102802	CONTINGENCY 'DVP_P4-2: 102802' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P4-6: CAROLIN T122	CONTINGENCY 'DVP_P4-6: CAROLIN T122' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 315126 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP2 14.400 OPEN BRANCH FROM BUS 314559 TO BUS 315128 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP4 14.400 OPEN BUS 315126 /* ISLAND: 1ROARAP2 14.400 OPEN BUS 315128 /* ISLAND: 1ROARAP4 14.400 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314578 CKT 1 /* 3CAROLNA 115.00 - 3HORNRTN 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 OPEN BUS 314559 /* 3CAROLNA 115.00 KV END
314906 8CLOVER 500 945810 AF1-246 TAP 500 1-A	CONTINGENCY '314906 8CLOVER 500 945810 AF1-246 TAP 500 1-A' OPEN BRANCH FROM BUS 314906 TO BUS 945810 CKT 1 END
DVP_P1-2: LN 556-B	CONTINGENCY 'DVP_P1-2: LN 556-B' OPEN BRANCH FROM BUS 945810 TO BUS 314936 CKT 1 /* AF1-246 TAP - 8RAWLINGS 500.00 END

Contingency Name	Contingency Definition
DVP_P7-1: LN 22-90-A	CONTINGENCY 'DVP_P7-1: LN 22-90-A' OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 /* 3EATON F 115.00 - AC1-054 TAP 115.00 OPEN BUS 314571 /* ISLAND: 3EATON F 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 /* 3FIVEFORKSDP115.00 - 3LITTLTN 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 /* 3FIVEFORKSDP115.00 - 3PALMERSPRNG115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 /* 3LITTLTN 115.00 - 3L GASTN 115.00 OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00 OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00 OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00 END
DVP_P4-2: 511T556	CONTINGENCY 'DVP_P4-2: 511T556' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 940470 TO BUS 314936 CKT 1 /* AE2-031 TAP 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 945810 TO BUS 314936 CKT 1 /* AF1-246 TAP 500.00 - 8RAWLINGS 500.00 END
DVP_P1-2: LN 98-C	CONTINGENCY 'DVP_P1-2: LN 98-C' OPEN BRANCH FROM BUS 924160 TO BUS 314267 CKT 1 /* AB2-060 TAP 115.00 - 3CHASCTY2 115.00 END
DVP_P7-1: LN 22-90-B	CONTINGENCY 'DVP_P7-1: LN 22-90-B' OPEN BRANCH FROM BUS 925780 TO BUS 314702 CKT 1 /* AC1-054 TAP 115.00 - 3KERR 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 /* 3FIVEFORKSDP115.00 - 3LITTLTN 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 /* 3FIVEFORKSDP115.00 - 3PALMERSPRNG115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 /* 3LITTLTN 115.00 - 3L GASTN 115.00 OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00 OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00 OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 2027-B	CONTINGENCY 'DVP_P1-2: LN 2027-B' OPEN BRANCH FROM BUS 933500 TO BUS 314333 CKT 1 /* AC2-165 TAP 230.00 - 6POWHATN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314322 CKT 1 /* 6JUDES F 230.00 - 6MDLTHAN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314333 CKT 1 /* 6JUDES F 230.00 - 6POWHATN 230.00 OPEN BUS 314310 /* ISLAND: 6JUDES F 230.00 OPEN BUS 314333 /* ISLAND: 6POWHATN 230.00 OPEN BUS 923862 /* ISLAND: AB2-026 E 230.00 END
DVP_P1-2: LN 1012-B	CONTINGENCY 'DVP_P1-2: LN 1012-B' OPEN BRANCH FROM BUS 936480 TO BUS 314681 CKT 1 /* AD2-063 TAP 115.00 - 3CHASCTY 115.00 END
DVP_P4-2: 101002	CONTINGENCY 'DVP_P4-2: 101002' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 313723 TO BUS 314259 CKT 1 /* 3PECAN 115.00 - 3CAROL56_1 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314835 CKT 1 /* 3CAROL56_1 115.00 - 3CAROL_1 115.00 OPEN BUS 314259 /* ISLAND: 3CAROL56_1 115.00 OPEN BUS 314835 /* ISLAND: 3CAROL_1 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P4-2: 2202	CONTINGENCY 'DVP_P4-2: 2202' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 /* 3EATON F 115.00 - AC1-054 TAP 115.00 OPEN BUS 314571 /* ISLAND: 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END

Contingency Name	Contingency Definition
Base Case	
DVP_P1-2: LN 296-A	CONTINGENCY 'DVP_P1-2: LN 296-A' OPEN BRANCH FROM BUS 304070 TO BUS 927250 CKT 1 /* 6PERSON230 T230.00 - AC1-221 TAP 230.00 END
DVP_P1-2: LN 296-B	CONTINGENCY 'DVP_P1-2: LN 296-B' OPEN BRANCH FROM BUS 927250 TO BUS 314697 CKT 1 /* AC1-221 TAP T230.00 - 6SEDGE HILL 230.00 END
DVP_P4-2: 13002	CONTINGENCY 'DVP_P4-2: 13002' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314595 TO BUS 314600 CKT 1 /* 3PL HILL 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314600 TO BUS 314615 CKT 1 /* 3PLHITP 115.00 - 3SKIPERS 115.00 OPEN BUS 314595 /* ISLAND: 3PL HILL 115.00 OPEN BUS 314600 /* ISLAND: 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P4-2: 556T591-A	CONTINGENCY 'DVP_P4-2: 556T591-A' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 945810 TO BUS 314936 CKT 1 /* AF1-246 TAP 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 314935 TO BUS 314936 CKT 1 /* 8HERITAGE 500.00 - 8RAWLINGS 500.00 END
DVP_P4-2: 511T591	CONTINGENCY 'DVP_P4-2: 511T591' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 940470 TO BUS 314936 CKT 1 /* AE2-031 TAP 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 314935 TO BUS 314936 CKT 1 /* 8HERITAGE 500.00 - 8RAWLINGS 500.00 END

12 Light Load Analysis

Not required for solar projects.

13 Short Circuit Analysis

The following Breakers are overdutied

None

13.1 System Reinforcements - Short Circuit

None

14 Stability and Reactive Power

Stability and Reactive Power requirement to be determined in the Facilities Study Phase.

15 Affected Systems

15.1 TVA

None

15.2 Duke Energy Progress

Customer may need to enter into an affected systems study with DEP. This will be confirmed in the Facilities Study phase.

16 Attachment 1: One Line Diagram

